

Explore Advanced Set Theory Concepts through Maxima

Maxima is a powerful free and open source Computer Algebraic System (CAS) that is capable of combining symbolic, numerical and graphical entities. This is the 19th article in our mathematical journey through open source, in which we explore advanced set theory concepts through Maxima.



With the introduction to set theory fundamentals in the previous article in this series, we are all set to explore the advanced realms of set theory through Maxima.

More set operations

We have already worked out the basic set creation techniques and some basic set operations provided by Maxima. Here are some next-level set operations it provides:

- makeset(expr, vars, varlist)** – Sophisticated set-creation using expressions
- adjoin(x, S)** – Returns a set with all elements of S and the element x
- disjoin(x, S)** – Returns a set with all elements S but without element x
- powerset(S)** – Returns the set of all subsets of S
- subset(S, p)** – Returns the subset of S, elements of which satisfy the predicate p
- symmdifference(S1, S2)** – Returns the symmetric difference between the sets S1 and S2, i.e., the elements in S1 or S2 but not in both

And here is a demonstration of each one of these operations:

```
$ maxima -q
(%i1) makeset(a+b, [a, b], [[1, 2], [2, 3], [3, 4], [4, 5], [5, 6]]);
(%o1) {3, 5, 7, 9, 11}
(%i2) makeset(a-b, [a, b], [[1, 2], [2, 3], [3, 4], [4, 5], [5, 6]]);
(%o2) {- 1}
(%i3) makeset(a*b, [a, b], [[1, 2], [2, 3], [3, 4], [4, 5], [5, 6]]);
(%o3) {2, 6, 12, 20, 30}
(%i4) makeset(a + 2*a*b + b, [a, b], [[1, 2], [2, 3], [3, 4], [4, 5], [5, 6]]);
(%o4) {7, 17, 31, 49, 71}
(%i5) quit();

$ maxima -q
(%i1) S: {-4, 6, 7, 32, 0};
(%o1) {- 4, 0, 6, 7, 32}
(%i2) adjoin(3, S);
(%o2) {- 4, 0, 3, 6, 7, 32}
(%i3) adjoin(7, S);
```